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12 January 1972
Superseding
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10 May 1966

MILITARY SPECIFICATION

EXERCISE HEAD TEST SET

MARK 485 MOD 0

(For Torpedo MK 46 MOD 0/1)

This Specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers requirements for the procurement of Exercise Head Test Set Mark 485 MOD 0, herein referred to as the "test set." (See 6.1 for description and intended use.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636 Boxes, Fiber

FSC 1355

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Military

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	Electronic Equipment, Naval Ship and Shore, General Specification

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibration of Shipboard Equipment

DRAWINGS

Naval Ordnance Systems Command

DL-2541000	Exercise Head Test Set, and all documents listed thereon.
LD-546960	Test Set, Exploder Mechanism Torpedo, Mk 440 Mod 0

PUBLICATIONS

Naval Ordnance Systems Command

WD 28955	Special Manufacturing Procedure for Calibration of the Exercise Head Test Set

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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PUBLICATIONS

Consolidated Classification Committee

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois. 60606)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall:

(a) Be manufactured using procedures proposed for the production lot.

(b) Consist of two exercise head test sets.

(c) One exercise head test set to be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the supplier's production methods are capable of yielding items that comply with technical requirements of the contract.

(d) The second exercise head test set to be supplied to an activity designated by the procuring activity for the purpose of performance verification.

3.2 Basic requirements

3.2.1 Contractor-furnished data. No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in manufacture of the exercise head test set shall be in accordance with requirements specified herein and with the applicable documents listed in NAVORD DL 2541000.

3.3 Performance requirements and product characteristics

3.3.1 Input power. The test set shall be capable of meeting the performance requirements specified herein when 115.0 ±11.5 volts alternating current (vac), at 60 ±3 Hertz (Hz), is supplied to connector pins J1-A and J1-B. The test set power source shall be capable of providing 300 watts continuously.

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3.3.2 Power circuits

3.3.2.1 Input. 115 vac (3.3.1) shall be provided to energize the +40 volt direct current (vdc) and the -40 vdc power supplies. The TEST SET POWER indicator lamp (red) shall illuminate when TEST SET POWER switch is placed to ON. The +40 vdc or -40 vdc power supplies shall provide all dc power required for test set functions.

3.3.2.2 Outputs. The dc voltages shall be connected to outputs of the test set. EXER HD POWER indicator lamp shall illuminate only when the EXER HD POWER switch is ON.

3.3.2.3 Oscillograph lamp power. An OSCILLOGRAPH LAMP switch shall provide +40.0 ±2.0 vdc to OSCILLOGRAPH LAMP output in ON position.

3.3.2.4 Oscillograph motor power. An OSCILLOGRAPH MOTOR switch shall provide -40.0 ±2.0 vdc to OSCILLOGRAPH MOTOR output in ON position.

3.3.3 Internally supplied voltages

3.3.3.1 Direct current voltages. The test set shall supply dc voltages of +40 vdc and -40 vdc. The dc voltages shall be regulated within ±0.2 vdc from no load to maximum load (40 ohms).

3.3.3.2 Alternating current voltages. The test set shall supply ac voltages of 5 volts root mean square (vrms), 26 vrms, and 32 vrms, all at a frequency of 1850 ±20 Hz. The 5 vrms and 26 vrms shall be regulated within ±5 percent from no load to maximum load (4700 ohms). The 32 vrms shall be regulated within ±4 vrms from no load to full load.

3.3.4 Test functions

3.3.4.1 Voltage monitoring. The VOLTAGE MONITOR meter shall monitor internal voltages when the EXER HD POWER switch is in the OFF position and external voltages when the EXER HD POWER switch is in the ON position. Voltage monitor indications shall be in accordance with Table I.

Table I

Voltage Monitor Indications

VOLTAGE SELECT Switch Position	Voltage Source	Meter Indication
+40	Internal	40 on 50 volt scale
	External	40 on 50 volt scale
-40	Internal	40 on 50 volt scale

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Table I (Cont'd)

Voltage Monitor Indications

VOLTAGE SELECT Switch Position	Voltage Source	Meter Indication
	External	40 on 50 volt scale
-10	External	10 on 15 volt scale
BAT. 1	External	2.65 on 5.0 volt scale
BAT. 2	External	2.65 on 5.0 volt scale

3.3.4.2 Oscillograph test circuitry

3.3.4.2.1 Oscillograph signal selection. Sequence of switching and signal outputs shall be as specified in Table II with COURSE GYRO switch positioned as indicated.

Table II

Oscillograph Input Switching Sequence

OSCILLOGRAPH GALVANOMETER INPUT SELECT Switch Position	Simulated Signal	Signal Voltage
1A	VALID ECHO	-12.0 ±0.6 vdc
1B	COURSE GYRO (46-0/1)	5.00 ±0.25 vrms
	COURSE GYRO (46-2)	+0.385 ±0.023 vdc
2A	RPM IND MOD "0"	-5.00 ±0.25 vdc*
2B	RPM IND MOD "1"	32.0 ±4.0 vrms
3A	RANGE GATE	-12.0 ±0.6 vdc
3B	AMPLITUDE THRESHOLD	-12.0 ±0.6 vdc
5A	RIGHT COMMAND	+5.00 ±0.25 vdc
5B	LEFT COMMAND	-5.00 ±0.25 vdc

*No load only-isolated through 15,000 ohms resistance

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Table II (Cont'd)

Oscillograph Input Switching Sequence

OSCILLOGRAPH GALVANOMETER INPUT SELECT Switch Position	Simulated Signal	Signal Voltage
5C	+K	-12.0 ±0.6 vdc
6A	DOWN COMMAND	-5.00 ±0.25 vdc
6B	UP COMMAND	+5.00 ±0.25 vdc
6C	-K	-12.0 ±0.6 vdc
6D	VALID ECHO INHIBIT	-12.0 ±0.6 vdc
7A	+90 V INST (SRA)	+40.0 ±2.0 vdc
7B	+90 V INST (INERTIA)	+40.0 ±2.0 vdc
7C	EXPLODER ARM	-40.0 ±2.0 vdc
8A	TARGET ANGLE THRESHOLD	-12.0 ±0.6 vdc
8B	NOISE THRESHOLD	-12.0 ±0.6 vdc

3.3.4.2.2 Oscillograph signal interrupt. A momentary switch (SIGNAL INTERRUPT) shall be provided, capable of opening the circuit through the OSCILLOGRAPH GALVANOMETER INPUT SELECT switch and interrupting the signal to the oscillograph.

3.3.4.2.3 Oscillograph signal test points. Jacks shall be provided on the test set panel which shall provide the capability to monitor the signal selected by the OSCILLOGRAPH GALVANOMETER INPUT SELECT switch. The jacks shall be placarded SIG and GRD.

3.3.4.3 Steering control test circuitry

3.3.4.3.1 Steering control signals. The test set shall provide steering control signals to the exercise head. With the STEERING switch in the NORMAL position, a -12.0 ±0.6 vdc signal shall be provided to (RANGE > B) and with the STEERING switch in the TURNAWAY position, a ground signal shall be provided to (RANGE > B).

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3.3.4.3.2 Steering control indications. The steering control circuitry shall provide an 1850 ±20 Hz, 20.5 ±2.5 volt peak-to-peak (p-p), square wave signal to ELECTRONIC DRIVE OUT. When a signal in-phase with the ELECTRONIC DRIVE OUT signal is applied to ELECTRONIC DRIVE IN INPUT, the STEERING-NORMAL indicator lamp shall illuminate. When a signal 180 degrees out-of-phase with the ELECTRONIC DRIVE OUT signal is applied to ELECTRONIC DRIVE IN INPUT, the STEERING-TURNAWAY indicator lamp shall illuminate.

3.3.4.4 Squib fire test circuitry.

3.3.4.4.1 Power failure squib. An indicator lamp (PWR FAILURE) shall be provided on the test set panel. The PWR FAILURE indicator lamp shall illuminate when a +40, +0 -20, vdc signal is applied to the PWR FAIL SQUIB input of the test set.

3.3.4.4.2 DCO squib. An indicator lamp (DCO) shall be provided on the test set panel. The DCO indicator lamp shall illuminate when a 2.7 ±0.3 vdc signal is applied across two DCO SQUIB inputs of the test set.

3.3.4.4.3 Termination squib. An indicator lamp (TERMINATION) shall be provided on the test set panel. The TERMINATION indicator lamp shall illuminate when a 2.7 vdc signal is applied across the TERM SQUIB input and the TERM SQUIB RET input of the test set.

3.3.4.5 Snake end indication. An indicator lamp (SNAKE END) shall be provided on the test set panel. The SNAKE END indicator lamp shall illuminate when a ground is provided at the 3 MIN SNAKE input of the test set.

3.3.4.6 Recovery indication. An indicator lamp (RECOVERY) shall be provided on the test set panel. The RECOVERY indicator lamp shall illuminate when a ground is provided at the RECOVERY SIGNAL input of the test set.

3.3.4.7 Inertia switch signal. A momentary switch shall be provided on the test set panel which, when depressed, shall provide a +40.0 ±2.0 vdc signal to the INERTIA SWITCH output of the test set.

3.3.4.8 Continuity and resistance. Resistance between the following pins of Connector J2 shall be as follows:

J2	Resistance
w to h (STEERING switch in TURNAWAY position)	8.0 to 8.4 K ohms
w to t	1.0 ohm max
w to v	1.0 ohm max
w to i	1.0 ohm max
w to K	1.0 ohm max
A to B	1.0 ohm max
A to C	1.0 ohm max

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3.3.4.8 Continuity and resistance. (Cont'd)

J2	Resistance
D to Z	1.0 ohm max
F to N	1.0 ohm max
J to L	1.0 ohm max
S to T } (See NOTE 1)	1.0 ohm max
V to U }	1.0 ohm max
a to TP 1 (GALVANOMETER INPUT SELECT switch in Pos 1-A)	1.0 ohm max
a to TP 1 (GALVANOMETER INPUT SELECT switch in Pos 1-A, S-2 depressed)	OPEN CIRCUIT

3.3.4.9 Arming plug interlock. An ARMING PLUG receptacle (J4805) shall be provided on the test set front panel. An exercise head test set arming plug (P4805) shall be provided and remain connected to test set ARMING PLUG receptacle (J4805) until an arming plug from an external source is to be connected to the receptacle.

3.3.4.10 Indicator colors. The indicator lamps used in the test set shall be colored as follows:

TEST SET POWER	Red
EXER HD POWER	Red
SNAKE END	Yellow
RECOVERY	Green
PWR FAILURE	Green
DCO	Green
TERMINATION	Green
NORMAL	Green
TURNAWAY	Red

NOTE 1 Arming plug connected to front panel

3.3.5 Service life. The service life of the test set shall be 5 years of normal service with a minimum of 60 days between calibrations.

3.4 Environmental requirements. Unless otherwise specified, the test set shall be capable of conforming to the performance and product characteristics of this specification, after being exposed, at any temperature in the applicable range, to the environmental requirements under all conditions specified in this section.

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3.4.1 Vibration. The test set shall be capable of withstanding the Type 1 vibration required by MIL-STD-167, except that the variable frequency and endurance test requirements shall be as follows: The test set in operating condition shall be vibrated from 5 to 15 Hz at a half peak amplitude of 0.030 ±0.006 inch and from 16 to 25 Hz at a half peak amplitude of 0.020 ±0.004 inch. The vibration shall be applied in discrete intervals of 1 Hz and applied for 1 minute at each interval.

3.4.2 Shock. The test set shall be capable of withstanding the shock required by MIL-S-901, for Grade B, Class I, lightweight equipment, Type A, in the nonoperating condition, except that the height of the hammer drops shall be; 1 foot, 1-1/2 feet, and 3 feet.

3.4.3 Inclination. The test set shall be capable of meeting the inclination requirements specified in MIL-E-16400.

3.4.4 Temperature. The test set shall be capable of withstanding the temperatures specified in MIL-E-16400 for Class 4 equipments.

3.4.5 Humidity. The test set shall be capable of withstanding the humidity requirements specified in MIL-E-16400.

3.5 Workmanship. The workmanship and finish shall be sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

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(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample, manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test set by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production, randomly select 1 unit from each group of 20 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

(b) Setp 2. Randomly select 1 unit from each group of 50 units. Continue with Setp 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

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(d) Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. The tests and examinations specified in 4.7 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees Fahrenheit (F))
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure

4.5 Preproduction tests. The test set submitted as a preproduction sample shall be subjected to all preproduction tests specified in 4.5.2. Prior to, and immediately after the preproduction tests, the test set shall be subjected to the quality conformance tests and examinations (see 4.7). Certified inspection data covering the results of preproduction tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial number of the test set.

(d) Test results of preproduction tests.

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein:

4.5.1.1 A vibration test fixture capable of applying vibration as specified in 3.4.1.

4.5.1.2 A shock fixture capable of applying shocks specified in 3.4.2.

4.5.1.3 An inclination testing device capable of performing the inclination tests specified in MIL-E-16400.

4.5.1.4 A humidity-temperature chamber capable of applying the humidity and temperature tests specified in MIL-E-16400.

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4.5.1.5 Items of test equipment specified in 4.7.2.

4.5.2 Preproduction tests

4.5.2.1 Vibration. Vibrate the test set in accordance with vibration schedules required by MIL-STD-167 with deviations specified in 3.4.1.

4.5.2.2 Shock. Secure the test set to the shock test fixture rigidly and apply shocks as specified in MIL-S-901 with deviations specified in 3.4.2.

4.5.2.3 Inclination. Subject the test set to the inclination tests specified in MIL-E-16400.

4.5.2.4 Temperature. Subject the test set to temperature tests, both operating and nonoperating, as specified in MIL-E-16400 for Class 4 equipments.

4.5.2.5 Humidity. Subject the test set to the humidily tests specified in MIL-E-16400.

4.6 Periodic production tests

4.6.1 Periodic production test equipment. The following items of test equipment, or equivalent items, will be required to perform the tests specified herein:

(a) Temperature chamber capable of providing the environment specified in 3.4.4.

(b) Vibration test fixture capable of applying vibration as specified in 3.4.1.

4.5.2 Periodic production tests. Each periodic production sample, after satisfactorily passing the quality conformance tests and examination (4.7), shall be subjected to environmental tests for Temperature (4.5.2.4) and for vibration (4.5.2.1). Unless otherwise specified, inability of the sample to pass satisfactorily the quality conformance tests and examination, subsequent to these environmental temperature and vibration tests, shall be deemed a defective unit.

4.7 Quality conformance tests and examinations. Unless otherwise specified, each test set delivered under contract shall be subjected to the tests and examinations specified herein.

4.7.1 Conditions

4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in order shown.

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4.7.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific indication obtained, the specific indication shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits column, the individual performing test or inspection shall initial in spaces provided in Reading column, indicating that requirements were satisfactorily met.

4.7.1.3 Quality conformance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command, via the cognizant Government representative.

4.7.2 Quality conformance test equipment. Items of test equipment listed in Table III are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

Table III

Quality Conformance Test Equipment

Item	Identification
Digital DC voltmeter (Range: 0.000 to 9.999 vdc; 00.00 to 99.99 vdc; Input impedance: 11 megohms; Accuracy: 0.2 percent of applied voltage or ±2 digits whichever is greater.)	Non-Linear Systems Model 451
Oscillograph (Bandpass: dc to 15 mega-Hertz; Rise time: 0.023 microseconds; Accuracy: 3 percent from 0.05 to 20 volts/cm.)	Tektronix Model 535A with 53/54 preamplifier
Electronic counter (Range: 10 Hz to 120 kiloHertz; Accuracy: ±1 count.)	Hewlett-Packard Model 522B
AC VTVM (Range: 1 millivolt to 300 vac; Accuracy: 1 percent at full scale; Frequency range: 10 Hz to 4 megaHertz).	Hewlett-Packard Model 400 H

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Table III (Cont'd)

Quality Conformance Test Equipment

Item	Identification
DC power supply (Regulated output: 0 to 30 vdc; Load regulation: less than 0.3 percent, or 30 millivolt change.)	Hewlett-Packard Model 721A
Multimeter (Resistance range: 0 to 20 megohms; Accuracy: ±3 percent of full scale.)	Simpson Model 260
Load Resistors	Commercial
40 ohms, 5%, 40 watts	
30,000 ohms, 5%, 1/2 watt	
4700 ohms, 5%, 1/2 watt	

4.7.3 Electrical tests

4.7.3.1 Initial test setup

4.7.3.1.1 Control settings. Verify arming plug is connected. Place the test set panel controls to the following initial positions.

Control	Setting
TEST SET POWER	OFF
EXER HD POWER	OFF
VOLTAGE SELECT	OFF
STEERING	TURNAWAY
OSCILLOGRAPH GALVANOMETER INPUT SELECT	OFF
LAMP	OFF
MOTOR	OFF

4.7.3.1.2 Continuity checks. Using multimeter, check for continuity and resistance between the following pins of connector J2. The resistance between each set of pins shall be as indicated.

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Multimeter Connection	Indication
w to h	8.0 to 8.4 K ohms
w to t	1.0 ohm max
w to v	1.0 ohm max
w to i	1.0 ohm max
w to K	1.0 ohm max
A to B	1.0 ohm max
A to C	1.0 ohm max
D to Z	1.0 ohm max
F to N	1.0 ohm max
J to L	1.0 ohm max
S to T	1.0 ohm max
V to U	1.0 ohm max
a to TP1 (GALVANOMETER INPUT SELECT switch in Pos 1-A)	1.0 ohm max
a to TP1 (GALVANOMETER INPUT SELECT switch in Pos 1-A, S-2 depressed)	OPEN CIRCUIT

4.7.3.2 Equipment turn-on

4.7.3.2.1 Connect test set cable W1 to AC POWER connector J1 on test set panel. Connect plug of cable W1 to 115 vac, 60 Hz power outlet.

4.7.3.2.2 Place TEST SET POWER switch on test set panel to ON. TEST SET POWER indicator on test set panel shall illuminate. Allow test set to warm up for 10 minutes.

4.7.3.3 Indicator lamp check

4.7.3.3.1 Depress each of the following indicator/switches on the test set panel. Each indicator shall illuminate when depressed.

SNAKE END	EXER HD POWER
PWR FAILURE	RECOVERY
TERMINATION	DCO
TURNAWAY	NORMAL

4.7.3.3.2 Verify that each indicator is colored as follows:

TEST SET POWER	red
EXER HD POWER	red
SNAKE END	yellow

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RECOVERY	green
PWR FAILURE	green
DCO	green
TERMINATION	green
NORMAL	green
TURNAWAY	red

4.7.3.4 +40 volt power supply test

4.7.3.4.1 Connect digital DC voltmeter to pins J2-w (GRD) and J2-z (+) on test set panel.

4.7.3.4.2 Place EXER HD POWER switch on test set panel to ON position. Digital DC voltmeter shall indicate +38.0 to +42.0 vdc.

4.7.3.4.3 Place EXER HD POWER switch to OFF position and connect 40-ohm load resistor across pins J2-w and J2-z.

4.7.3.4.4 Place EXER HD POWER switch to ON position. Digital DC voltmeter shall indicate +38.0 to +42.0 vdc.

4.7.3.4.5 Place EXER HD POWER switch to OFF position.

4.7.3.5 -40 volt power supply test

4.7.3.5.1 Connect digital DC voltmeter to pins J2-w (GRD) and J2-y (-) on test set panel.

4.7.3.5.2 Place EXER HD POWER switch on test set panel to ON position. Digital DC voltmeter shall indicate -38.0 to -42.0 vdc.

4.7.3.5.3 Place EXER HD POWER switch to OFF position and connect 40-ohm load resistor across pins J2-w and J2-y.

4.7.3.5.4 Place EXER HD POWER switch to ON position. Digital DC voltmeter shall indicate -38.0 to -42.0 vdc.

4.7.3.5.5 Place EXER HD POWER and TEST SET POWER switches to OFF position and disconnect digital DC voltmeter and load resistor.

4.7.3.6 1850 Hz power test

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4.7.3.6.1 Connect 30,000 ohm, 5 percent, load resistor across pins J2-AE and J2-K on test set panel to simulate load.

4.7.3.6.2 Connect AC VTVM between GRD test point on test set panel and pin J2-AE.

4.7.3.6.3 Place TEST SET POWER switch to ON position. AC VTVM shall indicate 24.7 to 27.3 vrms.

4.7.3.6.4 Connect electronic counter between GRD test point on test set panel and pin J2-AE. Indication shall be 1830 to 1870 Hz.

4.7.3.6.5 Place TEST SET POWER switch to OFF position and disconnect test equipment but not load resistor.

4.7.3.7 Voltage output test

4.7.3.7.1 Place TEST SET POWER switch and EXER HD POWER switch to ON positions. TEST SET POWER indicator shall illuminate.

4.7.3.7.2 Check voltage outputs of galvanometer input circuit test set by connecting voltmeter (AC or digital DC as indicated) between GRD test point on test set panel and pins of connector J2 as follows. Place OSCILLOGRAPH GALVANOMETER INPUT SELECT switch and COURSE GYRO switch in position indicated. Voltmeter shall indicate as specified in Table IV.

Table IV

Galvanometer Input Test

Test No.	Voltmeter Type	J2 Pin No.	Switch Position	Indication
1.	DC	a	1-A	-11.4 to -12.6 vdc
2.	AC	u	1-B (46-0/1)	4.75 to 5.25 vrms
3.	DC	u	1-B (46-2)	+0.362 to +0.408 vdc
4.	DC	b	2-A	-4.75 to -5.25 vdc
5.	AC	m	2-B	28.0 to 36.0 vrms*
6.	DC	c	3-A	-11.4 to -12.6 vdc
7.	DC	AL	3-B	-11.4 to -12.6 vdc
8.	DC	n	5-A	+4.75 to +5.25 vdc

*Measured with 4700 ohms load between pin J2-m and ground.

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Table IV (Cont'd)

Galvanometer Input Test

Test No.	Voltmeter Type	J2 Pin No.	Switch Position	Indication
9.	DC	p	5-B	-4.75 to -5.25 vdc
10.	DC	d	5-C	-11.4 to -12.6 vdc
11.	DC	q	6-A	-4.75 to -5.25 vdc
12.	DC	r	6-B	+4.75 to +5.25 vdc
13.	DC	f	6-C	-11.4 to -12.6 vdc
14.	DC	AM	6-D	-11.4 to -12.6 vdc
15.	DC	j	7-A	+38.0 to +42.0 vdc
16.	DC	k	7-B	+38.0 to +42.0 vdc
17.	DC	g	7-C	-38.0 to -42.0 vdc
18.	DC	AN	8-A	-11.4 to -12.6 vdc
19.	DC	AP	8-B	-11.4 to -12.6 vdc

4.7.3.7.3 With digital DC voltmeter connected as in last step, depress SIGNAL INTERRUPT switch on test set panel. Digital DC voltmeter shall indicate zero volts. When SIGNAL INTERRUPT switch is released, digital voltmeter indication shall return to -11.4 to -12.6 vdc.

4.7.3.7.4 Place OSCILLOGRAPH GALVANOMETER INPUT SELECT switch to OFF position.

4.7.3.7.5 Connect digital DC voltmeter between GRD test point on test set panel and pin J2-X.

4.7.3.7.6 Place LAMP switch to ON position. Digital DC voltmeter shall indicate +38.0 to +42.0 vdc. Place OSCILLOGRAPH LAMP switch to OFF position.

4.7.3.7.7 Connect digital DC voltmeter between GRD test point on test set panel and pin J2-W.

4.7.3.7.8 Place MOTOR switch to ON position. Digital voltmeter shall indicate -38.0 to -42.0 vdc. Place OSCILLOGRAPH MOTOR switch to OFF position.

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4.7.3.7.9 Connect digital DC voltmeter between GRD test point on test set panel and pin J2-h. Digital DC voltmeter shall indicate zero.

4.7.3.7.10 Place STEERING switch to NORMAL position. Digital DC voltmeter shall indicate -11.4 to -12.6 vdc.

4.7.3.7.11 Place EXER HD POWER switch to OFF position and disconnect test equipment.

4.7.3.8 Voltage monitor test

4.7.3.8.1 Place VOLTAGE SELECT switch on test set panel to +40 position. DC VOLTS meter on test set panel shall indicate +38.0 to +42.0 vdc. (Read on 50 volt scale.)

4.7.3.8.2 Place VOLTAGE SELECT switch on test set panel to -40 position. DC VOLTS meter on test set panel shall indicate -38.0 to -42.0 vdc. (Read indication on 50 volt scale.)

4.7.3.8.3 Place VOLTAGE SELECT switch on test set panel to OFF position and place EXER HD POWER switch on test set panel to ON position.

4.7.3.8.4 Connect jumper wire between pins J2-z and J2-V on test set panel.

4.7.3.8.5 Place VOLTAGE SELECT switch on test set panel to 240 position. DC VOLTS meter on test set panel shall indicate +38.0 to +42.0 vdc. (Read indication on 50 volt scale.) Place EXER HD POWER switch to OFF position.

4.7.3.8.6 Disconnect jumper wire from pins J2-z and J2-V, and connect between pins J2-y and J2-S on test set panel. Place EXER HD POWER switch to ON position.

4.7.3.8.7 Place VOLTAGE SELECT switch to -40 position. DC VOLTS meter on test set panel shall indicate -38.0 to -42.0 vdc. (Read indication on 50 volt scale.)

4.7.3.8.8 Connect external DC power supply outputs to digital DC voltmeter, set DC power supply SHORT switch to 225 MA and adjust DC power supply for an indication of 10.0 ±0.01 on digital DC voltmeter. Connect DC power supply to pins J2-w (+ lead) and J2-AD (- lead) on test set panel.

4.7.3.8.9 Place VOLTAGE SELECT switch on test set panel to -10 position. DC VOLTS meter on test set panel shall indicate -9.7 to -10.3 vdc. (Read indication on 15 volt scale.)

4.7.3.8.10 Connect external DC power supply outputs to digital DC voltmeter and adjust DC power supply for indication of 2.70 ±0.01 vdc. Connect DC power supply to pins J2-Y (+ lead) and J2-P (- lead) on test set panel.

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4.7.3.8.11 Place VOLTAGE SELECT switch on test set panel to BAT. 1 position. DC VOLTS meter on test set panel shall indicate +2.54 to 2.76 vdc. (Read indication on 5 volt scale.)

4.7.3.8.12 Connect leads of DC power supply to pins J2-Y (- lead) and J2-E (+ lead).

NOTE

DCO indicator shall illuminate during this step.

4.7.3.8.13 Place VOLTAGE SELECT switch on test set panel to BAT. 2 position. DC VOLTS meter on test set panel shall indicate -2.54 to -2.76 vdc. (Read indication on 5 volt scale.)

4.7.3.8.14 Place VOLTAGE SELECT switch on test set panel to OFF position, and disconnect and turn off DC power supply.

4.7.3.9 Indicator lamp test

4.7.3.9.1 Connect jumper wire between GRD test point on test set panel and pin J2-AA. SNAKE END indicator on test set panel shall illuminate.

4.7.3.9.2 Disconnect jumper wire from pin J2-AA and connect to pin J2-s. RECOVERY indicator on test set panel shall illuminate. Disconnect jumper wire.

4.7.3.9.3 Connect jumper wire between pins J2-z and J2-M. Turn exercise head power switch to ON. PWR FAILURE indicator on test set panel shall illuminate.

4.7.3.9.4 Disconnect jumper wire from pins J2-z and J2-M. PWR FAILURE indicator on test set panel shall extinguish.

4.7.3.9.5 Adjust output voltage of DC power supply to +2.7 volts DC and connect to pins J2-Y (+ lead) and J2-E (- lead). DCO indicator on test set panel shall illuminate.

4.7.3.9.6 Connect DC power supply to pins J2-P (+ lead) and J2-H (- lead). TERMINATION indicator on test set panel shall illuminate.

4.7.3.9.7 Disconnect and turn off DC power supply.

4.7.3.10 Steering test

4.7.3.10.1 Connect pin J2-AC on test set panel to channel A of oscilloscope (do not use probe) and input of electronic counter. Oscilloscope shall display a square wave with an amplitude of 18 to 23 volts p-p. Electronic counter shall display a frequency of 1830 to 1870 Hz.

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4.7.3.10.2 Set oscilloscope vertical scale to 0.2 VOLTS/CM and oscilloscope POLARITY switch to NORMAL position.

4.7.3.10.3 Connect vertical signal output of oscilloscope to pin J2-AB on test set panel. STEERING-NORMAL indicator on test set panel shall illuminate.

4.7.3.10.4 Set POLARITY switch on oscilloscope to INVERTED position. STEERING NORMAL indicator on test set panel shall extinguish and STEERING-TURNAWAY indicator on test set panel shall illuminate.

4.7.3.10.5 Place EXER HD POWER and TEST SET POWER switches to OFF position, disconnect oscilloscope from test set, and disconnect test set cable W1 from AC power outlet.

4.7.4 Examination of product. The test set shall be examined for conformance to the requirements specified herein and on the drawings listed in NAVORD LD 546960. The presence of any one of the following defects shall constitute a failure:

4.7.4.1 Test set not complete; components parts missing.

4.7.4.2 Component parts not new, evidence of reclaimed, used, or salvaged components.

4.7.4.3 Evidence of unauthorized materials or repairs.

4.7.4.4 Receptacles do not accept mating plugs.

4.7.4.5 Panel inscriptions incomplete, incorrect, or illegible.

4.7.4.6 Case not watertight.

4.7.4.7 Continuity not in accordance with schematic.

4.7.4.8 Test set not correctly identified.

4.7.4.9 Test set not preserved and packaged in accordance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A and Level C.

5.1.1.1 Cleaning and drying. The test set shall be thoroughly cleaned and dried by any process that will not damage the unit.

5.1.1.2 Packaging. The test set case shall serve as the package.

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5.1.2 Level B. Not applicable

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A. The test set, preserved and packaged in accordance with 5.1, shall be overpacked in containers in compliance with PPP-B-636 Type CF, Class-Weather Resistant, Variety SW, Grade V3C. (See 5.2.4.)

5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, when preserved and packaged as specified in 5.1, shall be overpacked to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.4. Cushing or dunnage. Supplier shall provide sufficient cushioning or dunnage material to adequately protect equipment from damage due to shock or vibration loads encountered during handling and transportation, and to assure safe delivery to the receiving activity.

5.3 Marking

5.3.1 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.2 Special marking. The following marking shall appear on the top and sides of each shipping container:

"DELICATE INSTRUMENT
HANDLE WITH CARE"

6. NOTES

6.1 Description and intended use

6.1.1 Description of test set. The test set is a portable unit of electronic test equipment contained in a watertight, shock protected combination case with a removable cover and facilities for storage of special tools and cabling. The case also provides protection from environmental deterioration while in storage and damage from handling during shipment. The test set front panel provides controls and connectors accessible for operation and interconnection of the test set when the cover is removed. The test set consists of electronic circuitry which provides proper simulated signals and loads to the exercise head unit being tested.

6.1.2 Intended use. The test set is to be used in the Fleet as electronic test equipment for checkout and test of Exercise Head Mk 85 Mod 0.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of packing.

(c) Assigned activity for preproduction and periodic tests.

(d) Required data.

(e) Any exceptions to this specification.

(f) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement given in one half the peak-to-peak excursion in inches or the corresponding maximum acceleration in gravity units (g). The frequency is given in Hertz (Hz).

6.3.2 Shock. Shock is an abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse force in units of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and a shock wave approximating a one-half sine wave. The time duration shall be measured between two points where two lines intersect the zero "g" line. These lines are tangent to the shock wave slopes.

6.3.3 Temperature. The temperature of the environment to which the equipment may be exposed, expressed in degrees Fahrenheit (°F).

6.3.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

EXERCISE HEAD TEST SET

DL 2541000

(For Torpedo Mk 46 Mod 0/1)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

EXERCISE HEAD TEST SET DL 2541000

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the test set bearing the recording equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test Sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the Test Set. Unless otherwise noted the sequence of testing, recording, and verification, as performed in 4.7, is mandatory and must be performed in the order shown.

20.2 Recording of Results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limits column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limits column, the individual performing the test or inspection shall initial in the space provided in the Reading column indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test equipment	4.7.2	In Calibration	_____
30.2 Test procedure	4.7.3		
30.2.1 Initial test setup	4.7.3.1		

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	Reference	Limits	Reading
30.2.1.1 Continuity and resistance between J2 connector pins	4.7.3.1.2		
w to h		8.0 to 8.4K ohms	_____
w to t		0 to 1.0 ohm	_____
w to v		0 to 1.0 ohm	_____
w to i		0 to 1.0 ohm	_____
w to K		0 to 1.0 ohm	_____
A to B		0 to 1.0 ohm	_____
A to C		0 to 1.0 ohm	_____
D to Z		0 to 1.0 ohm	_____
F to N		0 to 1.0 ohm	_____
J to L		0 to 1.0 ohm	_____
S to T		0 to 1.0 ohm	_____
V to U		0 to 1.0 ohm	_____
a to TP1 (GALVANOMETER INPUT SELECT switch in Pos 1-A)		0 to 1.0 ohm	_____
a to TP1 (GALVANOMETER INPUT SELECT switch in Pos 1-A, S-2 depressed)		OPEN CIRCUIT	_____
30.2.2 Equipment turn-on	4.7.3.2		
30.2.2.1 TEST SET POWER indicator	4.7.3.2.2	Illuminates	_____
30.2.3 Indicator lamp check	4.7.3.3		
30.2.3.1 Indicators illuminate when depressed	4.7.3.3.1		
EXER HD POWER (red)		Illuminates	_____
SNAKE END (yellow)		Illuminates	_____
RECOVERY (green)		Illuminates	_____
PWR FAILURE (green)		Illuminates	_____
DCO (green)		Illuminates	_____
TERMINATION (green)		Illuminates	_____
NORMAL (green)		Illuminates	_____
TURNAWAY (red)		Illuminates	_____
30.2.4 +40 volt power supply	4.7.3.4		
30.2.4.1 No load voltage	4.7.3.4.2	+38.0 to +42.0 vdc	_____

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	Reference	Limits	Reading
30.2.4.2 Voltage with load	4.7.3.4.4	+38.0 to +42.0 vdc	_____
30.2.5 -40 volt power supply	4.7.3.5		
30.2.5.1 No load voltage	4.7.3.5.2	-38.0 to -42.0 vdc	_____
30.2.5.2 Voltage with load	4.7.3.5.4	-38.0 to -42.0 vdc	_____
30.2.6 1850 Hz power	4.7.3.6		
30.2.6.1 Voltage	4.7.3.6.3	24.7 to 27.3 vrms	_____
30.2.6.2 Frequency	4.7.3.6.4	1830 to 1870 cps	_____
30.2.7 Voltage outputs	4.7.3.7		
30.2.7.1 TEST SET POWER indicator	4.7.3.7.1	Illuminates	_____
30.2.7.2 Galvanometer input circuit	4.7.3.7.2		
1.		-11.4 to -12.6 vdc	_____
2.		4.75 to 5.25 vrms	_____
		+0.362 to +0.408 vdc	_____
3.		-4.75 to -5.25 vdc	_____
4.		28.0 to 36.0 vrms	_____
5.		-11.4 to -12.6 vdc	_____
6.		-11.4 to -12.6 vdc	_____
7.		+4.75 to +5.25 vdc	_____

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30.2.7.2 (Cont'd)	Reference	Limits	Reading
8.		-4.75 to -5.25 vdc	_____
9.		-11.4 to -12.6 vdc	_____
10.		-4.75 to -5.25 vdc	_____
11.		+4.75 to +5.25 vdc	_____
12.		-11.4 to -12.6 vdc	_____
13.		-11.4 to -12.6 vdc	_____
14.		+38.0 to +42.0 vdc	_____
15.		+38.0 to +42.0 vdc	_____
16.		-38.0 to -42.0 vdc	_____
17.		-11.4 to -12.6 vdc	_____
18.		-11.4 to -12.6 vdc	_____
30.2.7.4 SIGNAL INTER-RUPT switch	4.7.3.7.3		
Depressed		-0.1 to +0.1 vdc	_____
Released		-11.4 to -12.6 vdc	_____
30.2.7.5 LAMP switch	4.7.3.7.6	+38.0 to +42.0 vdc	_____

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	Reference	Limits	Reading
30.2.7.6 MOTOR switch	4.7.3.7.8	-38.0 to -42.0 vdc	_____
30.2.7.7 STEERING switch to TURNAWAY	4.7.3.7.9	-0.1 to +0.1 vdc	_____
30.2.7.8 STEERING switch to NORMAL	4.7.3.7.10	-11.4 to -12.6 vdc	_____
30.2.8 Voltage monitoring	4.7.3.8		
30.2.8.1 Internal +40 volts	4.7.3.8.1	+38.0 to +42.0	_____
30.2.8.2 Internal -40 volts	4.7.3.8.2	-38.0 to -42.0	_____
30.2.8.3 External +40 volts	4.7.3.8.5	+38.0 to +42.0	_____
30.2.8.4 External -40 volts	4.7.3.8.7	-38.0 to -42.0	_____
30.2.8.5 External -10 volts	4.7.3.8.9	-9.7 to -10.3	_____
30.2.8.6 External BAT. 1	4.7.3.8.11	+2.54 to +2.76	_____
30.2.8.7 External BAT. 2	4.7.3.8.13	-2.54 to -2.76	_____
30.2.9 Indicator lamps	4.7.3.9		
30.2.9.1 SNAKE END indicator	4.7.3.9.1	Illuminates	_____
30.2.9.2 RECOVERY indicator	4.7.3.9.2	Illuminates	_____
30.2.9.3 PWR FAILURE indicator	4.7.3.9.3	Illuminates	_____
30.2.9.4 PWR FAILURE indicator	4.7.3.9.4	Extinguishes	_____
30.2.9.5 DCO indicator	4.7.3.9.5	Illuminates	_____
30.2.9.6 TERMINATION indicator	4.7.3.9.6	Illuminates	_____
30.2.10 Steering circuit	4.7.3.10		

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	Reference	Limits	Reading
30.2.10.1 ELECTRONIC DRIVE OUT signal	4.7.3.10.1		
Frequency		1830 to 1870 cps	_____
Amplitude		18 to 23 volts p-p	_____
30.2.10.2 NORMAL indicator	4.7.3.10.3	Illuminates	_____
30.2.10.3 TURNAWAY indicator	4.7.3.10.4	Illuminates	_____
30.3 Visual inspections	4.7.4		_____
30.3.1 Completeness	4.7.4.1		_____
30.3.2 Component parts	4.7.4.2		_____
30.3.3 Material and repairs	4.7.4.3		_____
30.3.4 Interchangeability	4.7.4.4		_____
30.3.5 Nomenclature, Control and identification callouts	4.7.4.5		_____
30.3.6 Watertightness	4.7.4.6		_____
30.3.7 Wiring	4.7.4.7		_____
30.3.8 Identification	4.7.4.8		_____
30.3.9 Preservation and packaging	4.7.4.9		_____

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-E-82447 (OS) REV. A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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